

ANALYZING AIR QUALITY INDEX AND POPULATION GROWTH DENSITY ACROSS U.S. REGIONS

presented by
Jyothi guttikonda

NASHVILLE SOFTWARE SCHOOL-
DDA16



WHAT IS AIR QUALITY?

Air quality refers to the condition of the air within our surroundings, determined by the presence of pollutants and their concentrations.

Maintaining good air quality is crucial for public health and the environment. Poor air quality can lead to various health issues, including respiratory diseases, cardiovascular problems, and even premature death.

In conclusion, air quality plays a vital role in environmental health and impacts us all. By monitoring and enhancing air quality, we can achieve improved health outcomes and contribute to a healthier planet.



WHAT IS AQI BASICS

Aqi(Air Quality Index) Is A Standardized System Used To Measure And Report Daily Air Quality.

The Environmental Protection Agency (Epa) Is The U.S. Government Agency Responsible For Protecting Human Health and the Environment.It Develops Air Quality Standards, Monitors Pollution Levels and Provides Aqi Data To The Public.

**PM2.5&
PM10**

Particulate Matter

S02

Sulfur Dioxide

N02

Nitrogen dioxide

C0

Carbon monoxide

OZONE

Ground-Level Ozone

WHY AQI MATTERS

- Provides insights into pollution levels and associated health risks.
- Offers guidance for public actions, such as outdoor activities and protective measures.

Air Quality Index (AQI) Values	Levels of Health Concern	Colors
<i>When the AQI is in this range:</i>	<i>...air quality conditions are:</i>	<i>...as symbolized by this color:</i>
0 - 50	Good	Green
51 - 100	Moderate	Yellow
101 - 150	Unhealthy for Sensitive Groups	Orange
151 - 200	Unhealthy	Red
201 - 300	Very Unhealthy	Purple
301 - 500	Hazardous	Maroon

MOTIVATION

Air pollution is increasingly becoming a significant issue in urban settings, directly affecting public health, climate, and overall quality of life. This capstone project aims to analyze the Air Quality Index (AQI) throughout Tennessee and its neighboring regions.

This area presents a unique opportunity to study air quality, as it encompasses both urban centers with rising populations and rural areas characterized by lower density.

OBJECTIVE

Analyze the Air Quality Index (AQI) across Tennessee and its neighboring states — Georgia, Alabama, Mississippi, Arkansas, North Carolina, South Carolina, Virginia, Missouri, and Kentucky from 2019 to 2024 to compare air quality patterns in relation to population and density.



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KEY TAKEAWAYS

- 2020 and early 2021 show lower AQI values — potentially influenced by reduced activity during the COVID-19 pandemic.

- *Summer is the most critical season for air quality management due to consistently higher AQI values.*
- *Winter tends to offer relatively cleaner air, although the degree of improvement varies.*

- High population density does not automatically mean poor air quality.
- Other factors such as industrial activity, geography, regulations, and transportation patterns likely play a significant role in AQI outcomes.

- *Despite being a highly populated city, Atlanta consistently shows relatively low AQI values, suggesting that factors beyond population size—such as regulations, geography, and urban planning—play a key role in maintaining better air quality.*

Recommendations for Improving Air Quality



Tracking seasonal trends helps policymakers and communities plan better pollution control strategies.

Air quality monitoring or interventions may be needed, especially in medium-sized cities.

Promote Urban Green Spaces in High-Density Areas.

Cities like Kansas City and Birmingham have notably high AQI despite lower density—investigate and address unique local pollution sources.



DATASOURCES

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Air Quality Data:

<https://files.airnowtech.org/?prefix=airnow>

Population Data:

<https://www.macrotrends.net/globalmetrics/cities/22922/atlanta/population>

Population Density Data:

<https://www.governing.com/archive/population-density-land-area-cities-map.html>



THANK YOU

“Turning Data into Insights for a Cleaner, Healthier Future.”

 **Q&A SESSION**